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# Identification of proteins and N-glycosylation sites of knee cartilage in Kashin-Beck disease compared with osteoarthritis

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ABSTRACT

The aim of this study was to identify crucial proteins and N-glycosylated sites in the pathological mechanism of Kashin-Beck disease (KBD) compared with osteoarthritis (OA). Nine KBD knee subjects and nine OA knee subjects were selected for the study. Quantitative proteomics and N-glycoproteomics data of KBD and OA were obtained by protein and N-glycoprotein enrichment and LC-MS/MS analysis. Differentially expressed proteins or N-glycosylation sites were examined with a comparative analysis between KBD and OA. Total 2205 proteins were identified in proteomic analysis, of which 375 were significantly different. Among these, 121 proteins were up-regulated and 254 were down-regulated. In N-glycoproteomic analysis, 278 different N-glycosylated sites that were related to 187 N-glycoproteins were identified. Proteins and their N-glycosylated sites are associated with KBD pathological process including ITGB1, LRP1, ANO6, COL1A1, MXRA5, DPP4, and CSPG4. CRLF1 and GLG1 are proposed to associate with both KBD and OA pathological processes. Key pathways in KBD vs. OA proteomic and N-glycoproteomic analysis contained extracellular matrix receptor interaction, focal adhesion, phagosome, protein digestion, and absorption. N-glycosylation may influence the pathological process by affecting the integrity of chondrocytes or cartilage. It regulated the intercellular signal transduction pathway, which contributes to cartilage destruction in KBD.

## 1. Introduction

Osteoarthritis (OA) is a progressive whole-joint disease characterized by abnormal subchondral bone perfusion and neovascular invasion into the synovium and articular cartilage [1]. It affects 7% of the global population, more than 500 million people worldwide [2]. In contrast to OA, Kashin-Beck disease (KBD) is an endemic, chronic deformative osteoarthropathy. It had a high prevalence and morbidity in Russia, North Korea, and China [3]. According to the statistical results in the year 2020 in China, there were about 104.26 million people at risk of KBD and about 0.17 million KBD patients [4]. The precisely etiological and pathological mechanisms of KBD remain unclear. Meanwhile, adult KBD patients are difficult to treat, and they will gradually develop similar symptoms to OA, for which a joint replacement is a method to significantly relieve pain [5]. Therefore, a comparative study between KBD and OA is very needed for the in vitro researches and clinical applications of KBD.

Studies have found that KBD and OA have similar pathological changes. These are degradation of extracellular matrix (ECM), reduction and disruption of proteoglycans (PGs) and collagens, and formation of cartilage lesions [6]. However, the two diseases are different in many aspects (Supplemental Table 1). The main pathological feature of KBD is a necrosis of the deep layer chondrocytes in the epiphyseal cartilage [7]. The pathological changes of OA are initially degeneration and inflammation of articular cartilage and then involve the entire joint. Cartilage integrity and maintenance are controlled by chondrocytes synthesizing and secreting mainly ECM macromolecules [8]. A comparative study on the expression and differences of the proteins of KBD and OA may help to well-understand the etiological and pathological mechanism of KBD.

Nowadays, proteomics is used to detect differential expression of

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proteins and their interactions in cells or tissues. However, protein functions are regulated by several factors, particularly protein post-translational modifications (PTMs) [9]. It has been found that glycosylation, especially N-glycosylation, is one of the major PTMs of proteins, which is also associated with OA [10]. Glycoproteomics is an important subdiscipline of proteomics, which aims to identify the changes in variable occupancy of different glycosylation sites and variable distribution of different glycans attached to a single glycosylation site [11,12]. However, there are quite a few studies on the comparison of proteomics and N-glycoproteomics of KBD vs. OA cartilage.

The present study aimed to find common differentially expressed N-glycoproteins and their N-glycosylation sites; common BP, CC, MF, and signaling pathways of KBD vs. OA in both proteomics and N-glycoproteomics. Our study was also focused on the N-glycosylation sites that were significantly different between KBD and OA. This study may help clarify the role of N-glycosylation and N-glycosylation sites in the pathological process of cartilage damage, and provide new ideas for the treatment of KBD and OA.

## 2. Materials and methods

### 2.1. Ethical permission, consent and cartilage harvest

Cartilage samples were collected from OA and KBD patients. All related analysis methods in the present study were approved by Ethics Committees at Xi'an Jiaotong University (No. 2017-196). All participants have signed informed consent for the study participation and publication of their clinical details and images before the study started.

### 2.2. Sample preparation

The scheme of the samples' preparation and analyses are presented in Fig. 1. Nine OA knee cartilage samples were collected from OA patients between 51 and 68 ($\bar{x} = 59.22$) years of age, who were from non-KBD-endemic areas from Xi'an, Shaanxi, China. OA was diagnosed based on the Western Ontario and McMaster Universities OA Index [13]. Nine KBD cartilage samples were collected from patients between 50 and 68 ($\bar{x} = 58.78$) years of age, who were living in the KBD-endemic areas of Linyou in the Shaanxi, China. KBD was diagnosed as grade I or II based on the clinical criteria (WST 207–2010) in China [14]. Detailed patient characteristics are shown in Supplemental Table 2.

After collection, articular cartilages were cut up by sterile scissors, snap-frozen for 5 min in liquid nitrogen, and stored at −80 °C before analysis. Three cartilage tissues from three different patients were mixed as one biological replicate. Then the protein extraction was performed as described previously [15]. In each group, 300 μg of the samples were weighed for filter-aided sample preparation enzymolysis for proteomics and N-glycoproteomics analysis, respectively [16]. The peptides of the samples were quantitated by measuring optical density at 280 nm.

### 2.3. Proteomics analysis

As described in the previous study [15], the samples (100 μg of each group) were used for Tandem Mass Tag (TMT) labeling as specified by the manufacturer's instructions (Thermo Fisher Scientific, Waltham, MA, USA). Then High-pH Reversed-Phase Fractionation in 1100 Series high-performance liquid chromatography (HPLC) Value System (Agilent Technologies, Waldbronn, Germany) equipped with a Gemini-NX column (00F-4453-E0, 4.6 × 150 mm, 3 μm, 110 Å, Phenomenex, Torrance, CA, USA) was used for separation of peptides. Then, quantitative proteomics analysis based on liquid chromatography-tandem mass spectrometry (LC-MS/MS) was applied for global proteomic analysis. Proteome Discoverer 2.1 (Thermo Fisher Scientific), with Matrix Science software built-in, was used for database identification and quantitative analysis against the Uniprot_human_156914_20170308.fasta.

### 2.4. N-glycoproteomics analysis

#### 2.4.1. N-glycopeptides enrichment and deglycosylation in $H_2^{18}O$

N-glycopeptides enrichment and deglycosylation were performed as

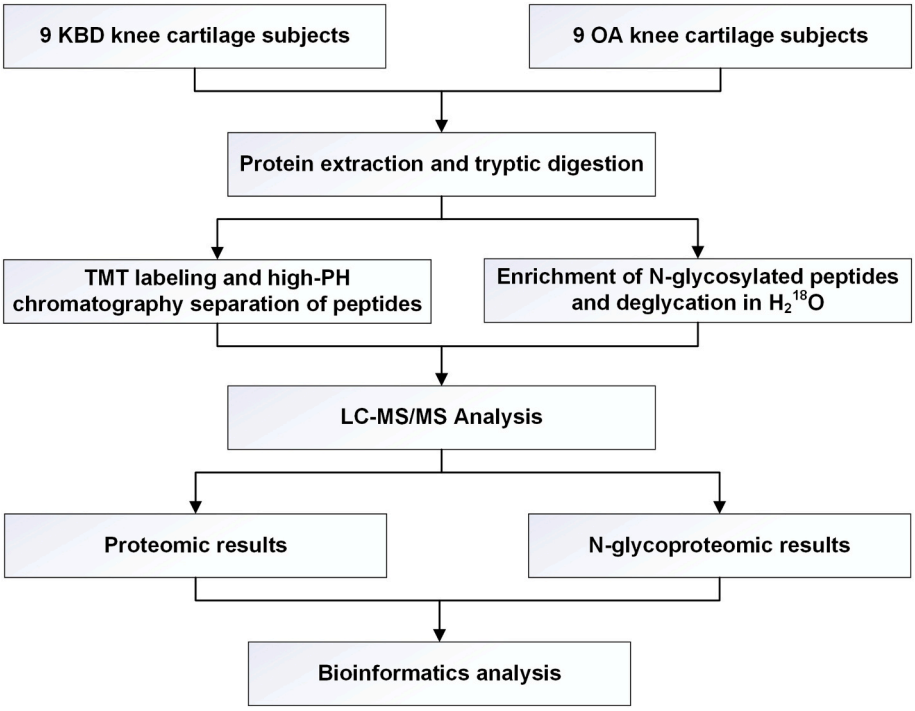



**Fig. 1.** Flow chart of the study. Nine knee cartilage samples were collected from KBD and OA patients, respectively. Proteins were extracted, denatured, and digested into peptides. Proteins and N-glycopeptides were enriched and then analyzed by LC-MS/MS. Finally, the functions of the identified proteins and N-glycoproteins were characterized by bioinformatics analysis.

previous studies described [17] with some modifications. The peptides (300 μg) were transferred to a new 30 kilodalton (kDa) ultrafiltration centrifuge tube, and then 50 μL of lectin mixture CWR (Concanavalin A, wheat germ agglutinin, and *Ricinus communis* agglutinin) was added. The resulting mixture was incubated for 1 h at room temperature after being shaken at 600 rpm for 1 min. And the unbound peptides were eluted by centrifugation. The lectin-bound peptides were washed four times with 200 μL 1× binding buffer and one time with 50 μL 25 mM NH4HCO3 in $H_2^{18}O$. The deglycosylation was carried out by adding 50 μL of 25 mM $NH_4HCO_3$ containing 2 μg of peptide N-glycosidase F and incubated at 37 °C for 3 h. Then the deglycosylated peptides were eluted with 40 μL 25 mM NH4HCO3 in $H_2^{18}O$, and lyophilized and reconstituted in 6 μL of 0.1% formic acid (FA).

#### 2.4.2. LC-MS/MS for N-glycoproteomic analysis

Easy-nLC1000 HPLC system (Thermo Scientific, USA) connected to Thermo Fisher Scientific Orbitrap Fusion Lumos mass spectrometer (Thermo Scientific, USA) was used to quantitative deglycosylated peptides. A total of 3 μg samples were loaded with an auto-sampler. The peptide traps Thermo scientific EASY column (2 cm * 100 μm 5 μm-C18) and Thermo scientific EASY column (75 μm * 100 mm 3 μm-C18) balanced by solvent A (0.1% formic acid in 2% acetonitrile) was used for peptide separation at a flow rate of 250 nL/min. A 60 min gradient starting at 2% and ending by 90% of solvent B (0.1% formic acid in 90% acetonitrile) was applied.

The mass spectrometer was operated in positive ion mode. MS spectra were acquired over a range of 350–1600 *m/z*. Master scan for full scans acquisition in the Orbitrap with 60,000 resolutions at *m/z* 200. For peptides and peptide fragments, they were detected by performing an MS2 scan. The resolving powers of the MS2 scan were 120 *m/z* for 15,000 resolutions. The maximum ion injection times were set to 50 ms for the MS scan and 22 ms for the MS2 scan. The automatic gain control target values for MS modes were set to 4e6, and 5e4 for MS2.

#### 2.4.3. Identification and quantification of intact N-glycopeptides

Raw data of mass spectrometry were analyzed using MaxQuant (1.6.5.0) software [18]. To obtain the modified glycoproteins, glycopeptides and glycosites, a total of 177661 sequences were compared with Uniprot-human_20190102_177661.fasta. The screening criteria were set as follows: methionine oxidation and asparagine deamidation were set up as variable modifications, and cysteine carbamidomethylation as a fixed modification. Trypsin and protein false discovery rate ≤ 0.01 were used as digestion mode, with four residues as the minimum length of the peptides. The unique peptide that only has N-X-S/T sequon was considered to be identified glycopeptide.

### 2.5. Bioinformatics analysis

Normalized glycopeptide spectral data sets were unit variance scaled and mean-centered, then analyzed by principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) using the MetaboAnalyst (version 5.0) [19]. Model quality was evaluated using $R^2Y$ and $Q^2$ values. PCA was conducted to investigate the homogeneity within groups and heterogeneity between groups. OPLS-DA was performed to distinguish the two groups.

Significantly differentially expressed proteins and N-glycoproteins were screened according to the standard of variation ratio ≥ 1.2 and *P*-value ≤ 0.05. GraphPad Prism 8 (GraphPad Software, San Diego, CA, USA) was used for the generation of a volcano map that was used to visualize up-regulated and down-regulated proteins and glycoproteins. Origin 2021 was used to create Venn diagrams that showed mathematical or logical connections between different sets intuitively.

The UniProt Knowledgebase (UniProtKB, https://www.uniprot.org) was used to collect detailed annotations extracted from the literature by expert curators [20]. Some of the glycoproteomic searches are from the glycoproteomic databases GlyGen Portal (https://www.glygen.org) [21].

Gene Ontology (GO, http://www.geneontology.org) annotation and Kyoto Encyclopedia of Genes and Genomes (KEGG http://www.kegg.jp/kegg/pathway.html) pathway enrichment analysis of the differential glycopeptides matched glycoproteins between KBD and OA were investigated using a database for annotation. The identified differentially expressed proteins in GO were annotated as three categories as molecular function (MF), biological process (BP), and cellular component (CC) based on the GO analysis. Fisher's Exact Test was used to identify the level of the enriched proteins in KEGG pathway enrichment analysis.

## 3. Results

### 3.1. Identification of proteins and N-glycosylation sites in KBD and OA

PCA and OPLS-DA analyses showed homogeneity and heterogeneity in and between KBD and OA. The unsupervised analysis of PCA provided a model with two components explaining 68.6% for proteomic results and 72.1% for N-glycoproteomics of the variance. No detection of outliers was found in the visual inspection of scores (Fig. 2A, B). OPLS-DA, a supervised analysis led to a good separation between groups with acceptable values of explained variation related to proteomic classes ($R^2Y = 0.94$) and predictive ability ($Q^2 = 0.73$) (Fig. 2C) and N-glycoproteomics classes ($R^2Y = 0.87$) and predictive ability ($Q^2 = 0.71$) (Fig. 2D).

A total of 2205 proteins were identified in both KBD and OA by proteomic analysis. Among these, 375 proteins were significantly different. Compared with OA cartilages, KBD ones had 121 significantly up-regulated proteins (38 proteins $P < 0.01$, 83 proteins $0.01 \leq P < 0.05$) and 254 significantly down-regulated proteins (50 proteins $P < 0.01$, 204 proteins $0.01 \leq P < 0.05$) (Fig. 2E). A total of 278 N-glycosylation sites were identified in KBD and OA cartilages with N-glycoproteomics analysis, among which 150 sites were significantly down-regulated (73 sites $P < 0.01$, 77 sites $0.01 \leq P < 0.05$) (Fig. 2F).

### 3.2. Differentially expressed proteins in proteomic and N-glycoproteomic analysis of KBD vs. OA

N-glycoproteomic analysis showed that 66 N-glycosylation sites were present in KBD cartilages (KBD unique), and defined as up-regulated sites. These sites were related to 61 proteins and defined as up-regulated proteins in N-glycoproteomics (Fig. 3A yellow part). Sixty-two N-glycosylation sites on 54 proteins were found only in OA cartilages (OA unique). There were 150 significantly down-regulated N-glycosylation sites found in KBD vs. OA related to 98 proteins. OA unique N-glycosylation sites and significantly down-regulated sites were jointly defined as 212 down-regulated N-glycosylation sites. They were related to 139 down-regulated proteins and defined as down-regulated proteins in N-glycoproteomics (Fig. 3A purple part).

The Venn diagram in Fig. 3A showed that the number of up-regulated proteins was 114 only in the proteomic analysis. The down-regulated proteins were 250. 45 up-regulated proteins and 119 down-regulated proteins were found only in N-glycoproteomic analysis. However, there were 12 proteins up-regulated and down-regulated at different N-glycosylation sites in N-glycoproteomic analysis. There was only one protein found to be down-regulated in proteomic analysis, but it was also up-regulated and down-regulated at a different site in N-glycoproteomic analysis. Seven proteins were up-regulated in proteomic analysis, but down-regulated in N-glycoproteomic analysis. Only three proteins were down-regulated in proteomic analysis but up-regulated in N-glycoproteomic analysis. The detailed information of the differentially expressed proteins in the opposite direction of proteomic and N-glycoproteomic analysis results was presented in Table 1.

A comparative analysis was performed to compare all proteins detected in proteomic analysis (2205 in total including non-significant

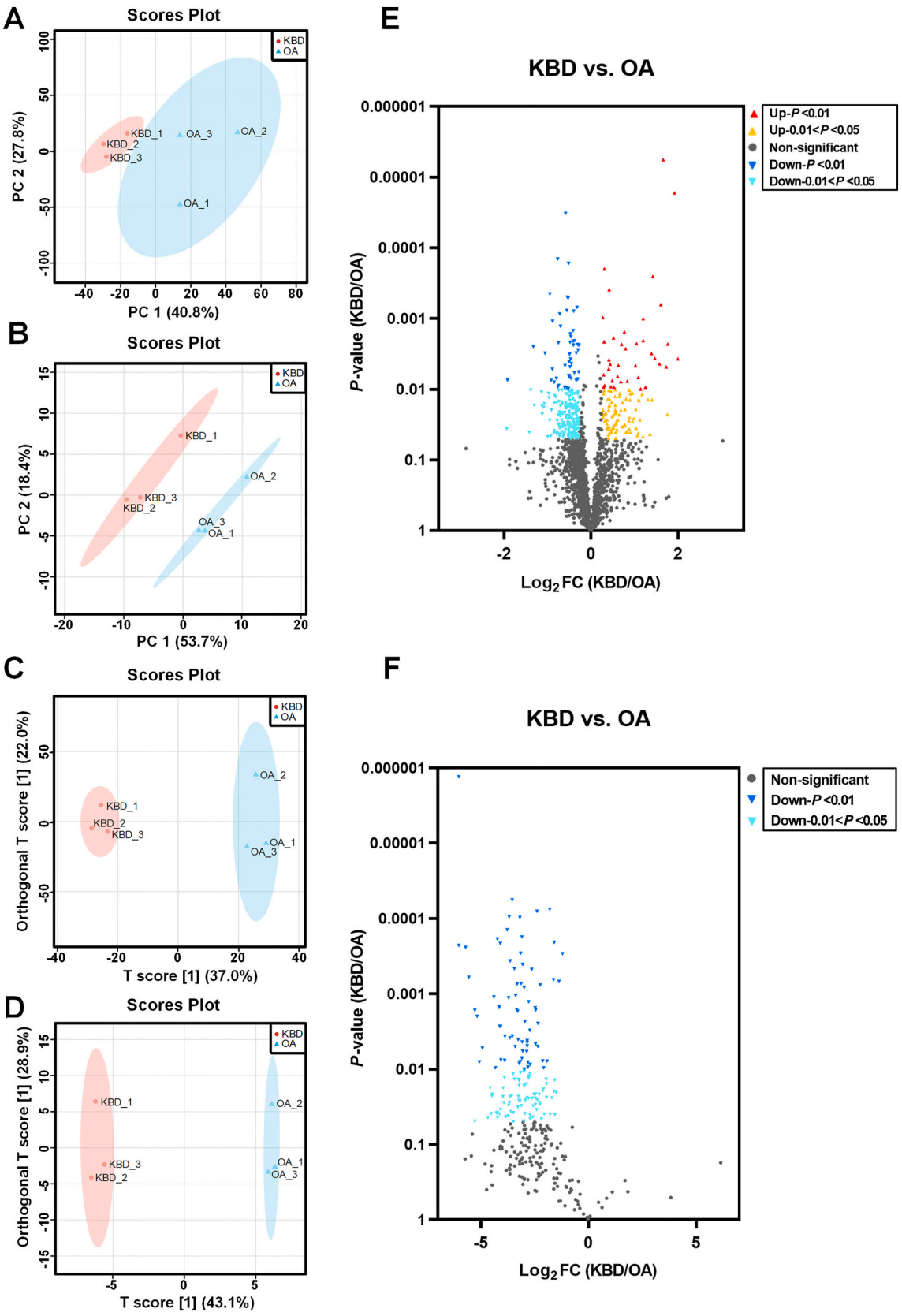



**Fig. 2.** Qualitative and quantitative analysis of the identified proteins and N-glycosylated peptides in KBD and OA cartilage. (A) Principal component analysis (PCA) for proteomic results of the samples in KBD and OA knee cartilage. (B) PCA for N-glycoproteomic results of the samples in KBD and OA knee cartilage. (C) Orthogonal partial least squares discriminant analysis (OPLS-DA) for proteomic results of the samples in the KBD and OA knee cartilage. (D) Differential expression of the proteins found in proteomic analysis in KBD vs. OA cartilage (E) Differentially expressed proteins in KBD vs. OA cartilage. (F) Differential expression of N-glyco-peptides from N-glycoproteomic analysis in KBD vs. OA cartilage.

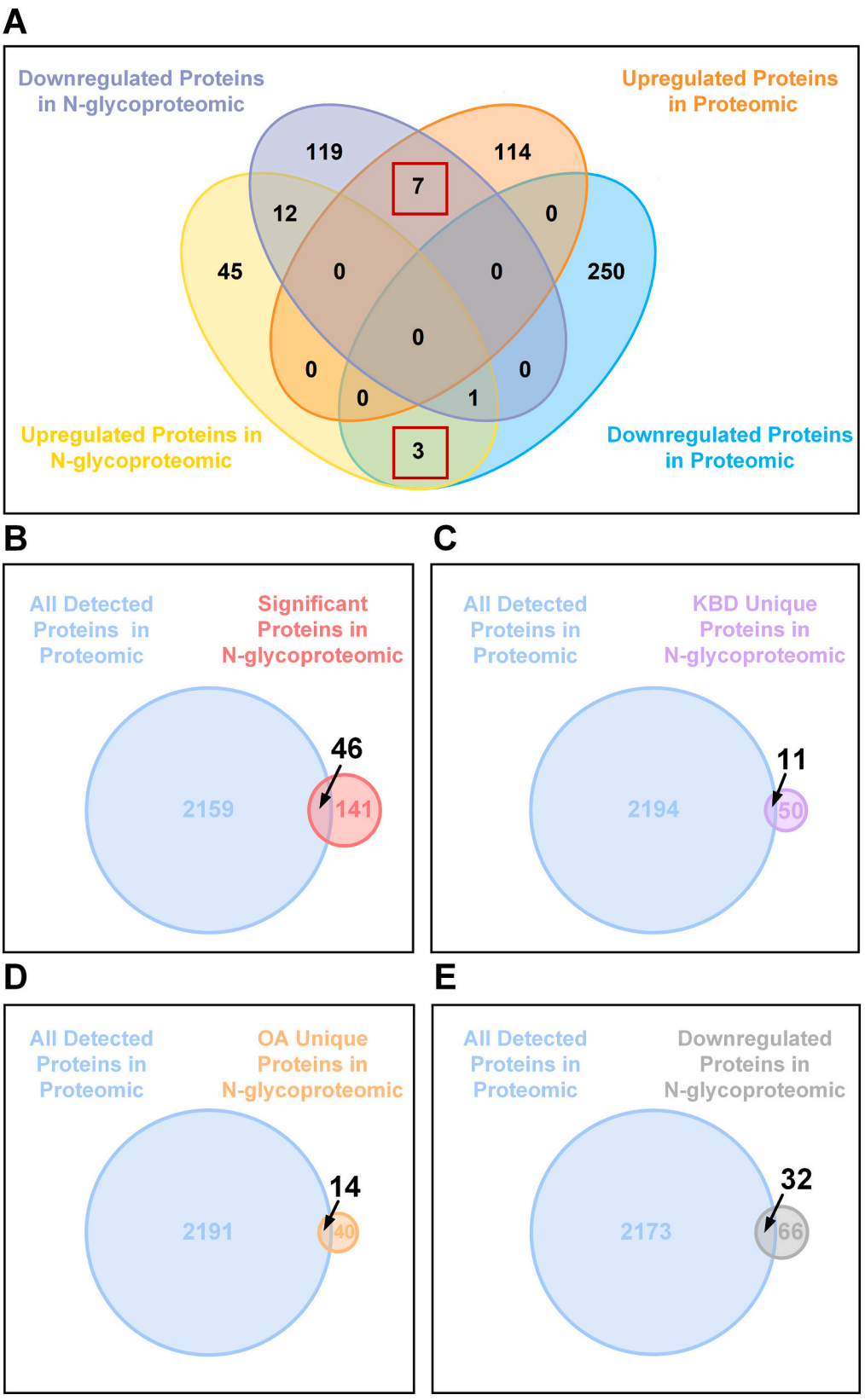



**Fig. 3.** Differentially expressed proteins in proteomic and N-glycoproteomic analysis of KBD and OA patients. (A) Venn diagram between up-regulated proteins and down-regulated proteins in proteomic and N-glycoproteomic analysis identified in the cartilage of KBD vs. OA patients. (B) Venn diagram between all detected proteins in proteomic analysis and significant proteins in N-glycoproteomic analysis. (C) Venn diagram between all detected proteins in proteomic analysis and KBD unique proteins in N-glycoproteomic analysis. (D) Venn diagram between all detected proteins in proteomic analysis and OA unique proteins in N-glycoproteomic analysis. (E) Venn diagram between all detected proteins in proteomic analysis and downregulated proteins in N-glycoproteomic analysis.

**Table 1**
Proteins differentially expressed in N-glycoproteomic and proteomic of KBD vs. OA.

| Group | Uniprot entry ID | Gene | Protein | KBD/OA protein abundance ratio | KBD/OA protein abundance ratio *P*-value | KBD/OA N-glycosylation-modified protein abundance ratio | KBD/OA N-glycosylation-modified protein abundance ratio *P*-value | Deamidation 18O (N) probabilities |
|---|---|---|---|---|---|---|---|---|
| Downregulated in proteomic and upregulated in N-glycoproteomic | D3DTX7 | COL1A1 | Collagen, type I, alpha 1, | 0.698 | 0.047 | KBD unique | – | LMSTEASQN(1)ITYHCK |
| | Q6UVK1 | CSPG4 | Chondroitin sulfate proteoglycan 4 | 0.796 | 0.011 | KBD unique | – | GVN(1)ASAVVN(1)VTVR |
| | P05556 | ITGB1 | Integrin beta-1 | 0.832 | 0.040 | KBD unique | – | CHEGN(1)GTFECGACR |
| | | | | | | KBD unique | – | KEN(1)SSEICSN(0.058)N(0.942) GECVCGQCVCR |
| | Q9UKU9 | ANGPTL2 | Angiopoietin-related protein 2 | 2.966 | 0.004 | 0.030 | 0.008 | ILN(1)QTADMLQLASK |
| | | | | | | 0.111 | 0.008 | YQHLATLAHN(1) QSEIIAQLEEHCQR |
| | A3KC71 | Okuribin | Nuclear envelope protein okuribin | 1.927 | 0.011 | 0.180 | 0.002 | ISVITIFGPVN(0.094)N(0.906) STMK |
| | | | | | | OA unique | – | LLGVGEEVGGVLELFPIN(1) GSSVVER |
| Upregulated in proteomic and downregulated in N-glycoproteomic | O75339 | CILP | Cartilage intermediate layer protein 1 | 1.841 | 0.010 | 0.026 | 0.002 | DAVQN(1)CCGISK |
| | | | | | | 0.059 | 0.003 | EQRPGQN(1)CSNYTVR |
| | | | | | | 0.028 | 0.002 | LPHDCFQN(1)ATN(1)SFYYDVGR |
| | | | | | | 0.015 | <0.001 | YFWYHN(1)DTLLDPSLYK |
| | | | | | | OA unique | – | DRDQPN(1)VSAACLEFK |
| | | | | | | OA unique | – | N(1) ISTATAAQTDLNFINDEGDTFPLR |
| | P23142 | FBLN1 | Fibulin-1 | 1.481 | 0.040 | 0.045 | 0.021 | CATPHGDN(1)ASLEATFVK |
| | O60687 | SRPX2 | Sushi repeat-containing protein | 1.322 | 0.020 | OA unique | – | INVN(1)VNSAAGLLDQFYEK |
| | P82987 | ADAMTSL3 | ADAMTS-like protein 3 | 1.356 | 0.004 | OA unique | – | AN(1)GTVQVVSPR |
| | Q8IWU5 | SULF2 | Extracellular sulfatase Sulf-2 | 1.629 | 0.002 | OA unique | – | LFPN(1)ASQHITPSYNYAPNPDK |
| | | | | | | OA unique | – | YVHNHNTYTNNEN(1) CSSPSWQAQHESR |

ones) with the proteins associated with significant N-glycosylation sites related proteins in N-glycoproteomic analysis (187 in total). 46 proteins were found to be shared (Fig. 3B). A total of 2205 proteins detected in proteomic analysis shared 11 proteins with 61 KBD unique N-glycosylation sites related proteins (Fig. 3C). It also shared 14 proteins when compared with 54 OA unique N-glycosylation sites related proteins (Fig. 3D). Additionally, 32 shared proteins were found in the comparison of all detected proteins in proteomics with 98 down-regulated N-glycosylation sites related proteins in N-glycoproteomic analysis (Fig. 3E). Detailed information was shown in Supplemental Table 3.

### 3.3. Common GO pathways in proteomic and N-glycoproteomic analysis of KBD vs. OA

GO terms and GO enrichment analysis was performed on the results of proteomic and N-glycoproteomic analysis. The top 20 GO significance terms are displayed through bubble charts in BP, CC, and MF, respectively. Among the top 20 enrichment analyses, results of BP, proteomics and N-glycoproteomics shared 8 GO terms that were mainly involved in the following BP: response to wounding, single-multicellular organism process, multicellular organismal process, multicellular organism development, system development, developmental process, anatomical structure development, single-organism developmental process (Fig. 4A, B). Among the top 20 CC enrichment results, there were 10 common GO terms: extracellular region part, extracellular organelle, extracellular membrane-bounded organelle, extracellular vesicle, extracellular exosome, membrane-bounded vesicle, vesicle, extracellular region, extracellular space, ECM (Fig. 4C, D). Among the top 20 results of MF enrichment analysis, there were two common GO terms: ECM structural constituent, calcium ion binding (Fig. 4E, F).

### 3.4. Common KEGG pathways in proteomic and N-glycoproteomic analysis of KBD vs. OA

KEGG pathways analysis showed that 22 KEGG pathways were significantly enriched in proteomic analysis. They belonged to metabolism, genetic information processing, environmental information processing, cellular processes, organismal systems, and human disease categories (Fig. 5A). Eleven KEGG pathways were significantly enriched in N-glycoproteomic analysis. They belonged to metabolism, environmental information processing, cellular processes, human disease, and unknown categories (Fig. 5B). Among significant results and the top 20 enriched KEGG pathways of proteomic and N-glycoproteomic analysis, four KEGG pathways were also shown in the Venn diagram ($P < 0.05$) (Fig. 5C). The common KEGG pathways were: ECM-receptor interaction, focal adhesion, phagosome, protein digestion and absorption.

## 4. Discussion

The present study in proteomic and N-glycoproteomic identified some N-glycoproteins and N-glycosylation sites in KBD and OA. It may help to deepen our understanding of cartilage degeneration through glycosylation modification. The proteins in N-glycosylation sites that might be significant for the pathogenesis of KBD were: integrin β1 (ITGB1), prolow-density lipoprotein receptor-related protein 1 (LRP1), anoctamin-6, collagen type I alpha1 chain (COL1A1), matrix-remodeling-associated protein 5 (MXRA5), golgi apparatus protein 1 (GLG1). The protein in N- glycosylation sites that might be significant for OA pathogenesis was chondroitin sulfate proteoglycan 4 (CSPG4). Dipeptidyl peptidase 4 (DPP4) has two different N- glycosylation sites in KBD and OA, respectively.

Integrin is a heterodimeric glycoprotein that consists of both an α and a β subunit [22]. ITGB1 can bind to 12 different α subunits (α1-11, αV)

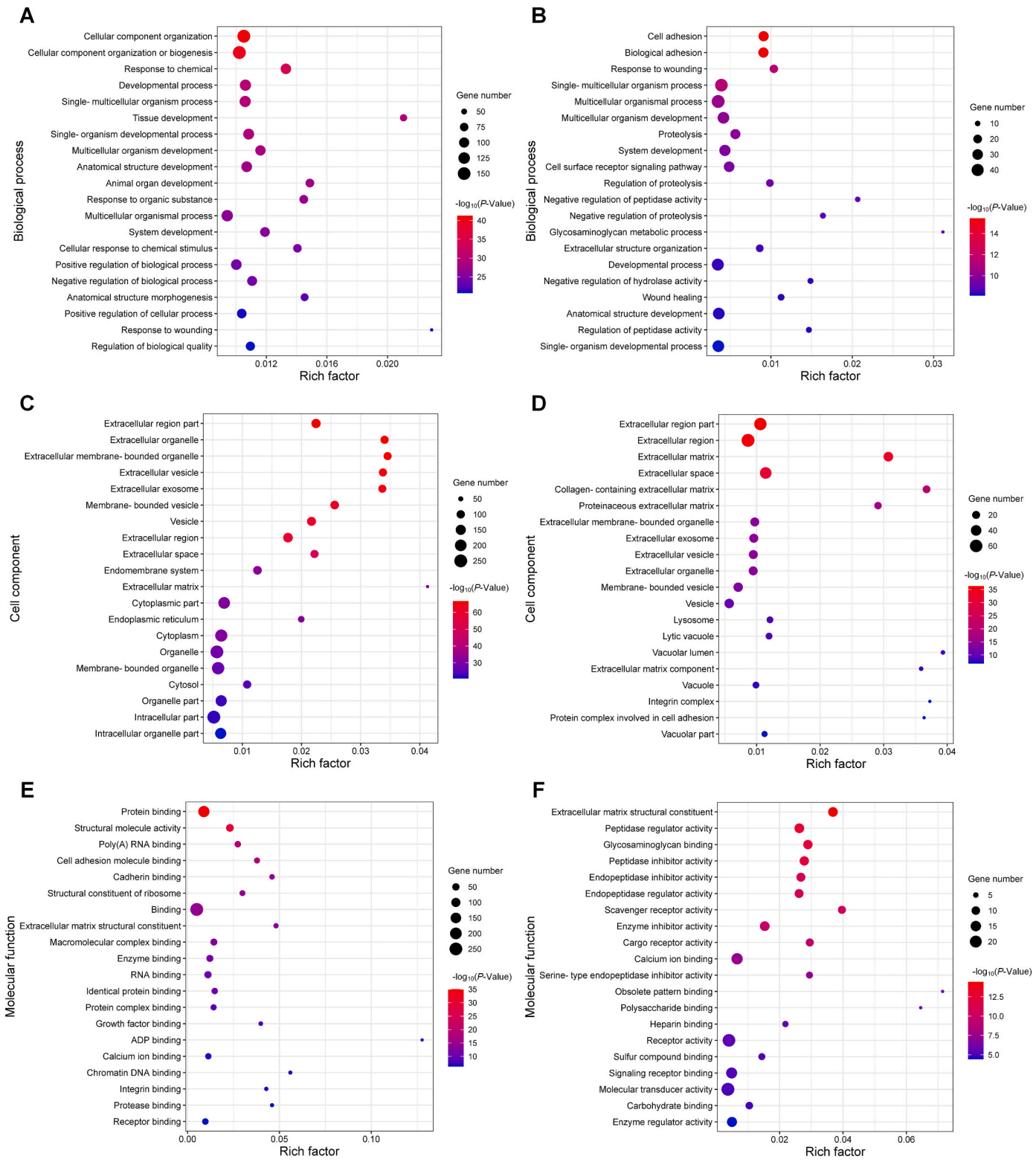



**Fig. 4.** Top 20 GO enrichment results of proteomic and N-glycoproteomic analysis in the cartilage of KBD and OA patients. (A) Top 20 biological process enrichment results of proteomic analysis. (B) Top 20 biological process enrichment results of N-glycoproteomic analysis. (C) Top 20 cell component enrichment results of proteomic analysis. (D) Top 20 cell component enrichment results of N-glycoproteomic analysis. (E) Top 20 molecular function enrichment results of proteomic analysis. (F) Top 20 molecular function enrichment results of N-glycoproteomic analysis.

thereby obtaining a variety of ligand specificities [22]. The specificity of many integrins is dependent on the alpha subunits and, thus influences the effects on the cellular behavior [23]. ITGB1 is critical for the structure, function and development of cartilage. It has been shown that deletion of ITGB1 in the limb bud of mutant mice resulted in multiple abnormalities of the knee joints [24]. ITGB1 overexpression may promote chondrogenic differentiation of mesenchymal stem cells [25]. Proinflammatory cytokines increased the expression of ITGB1 on

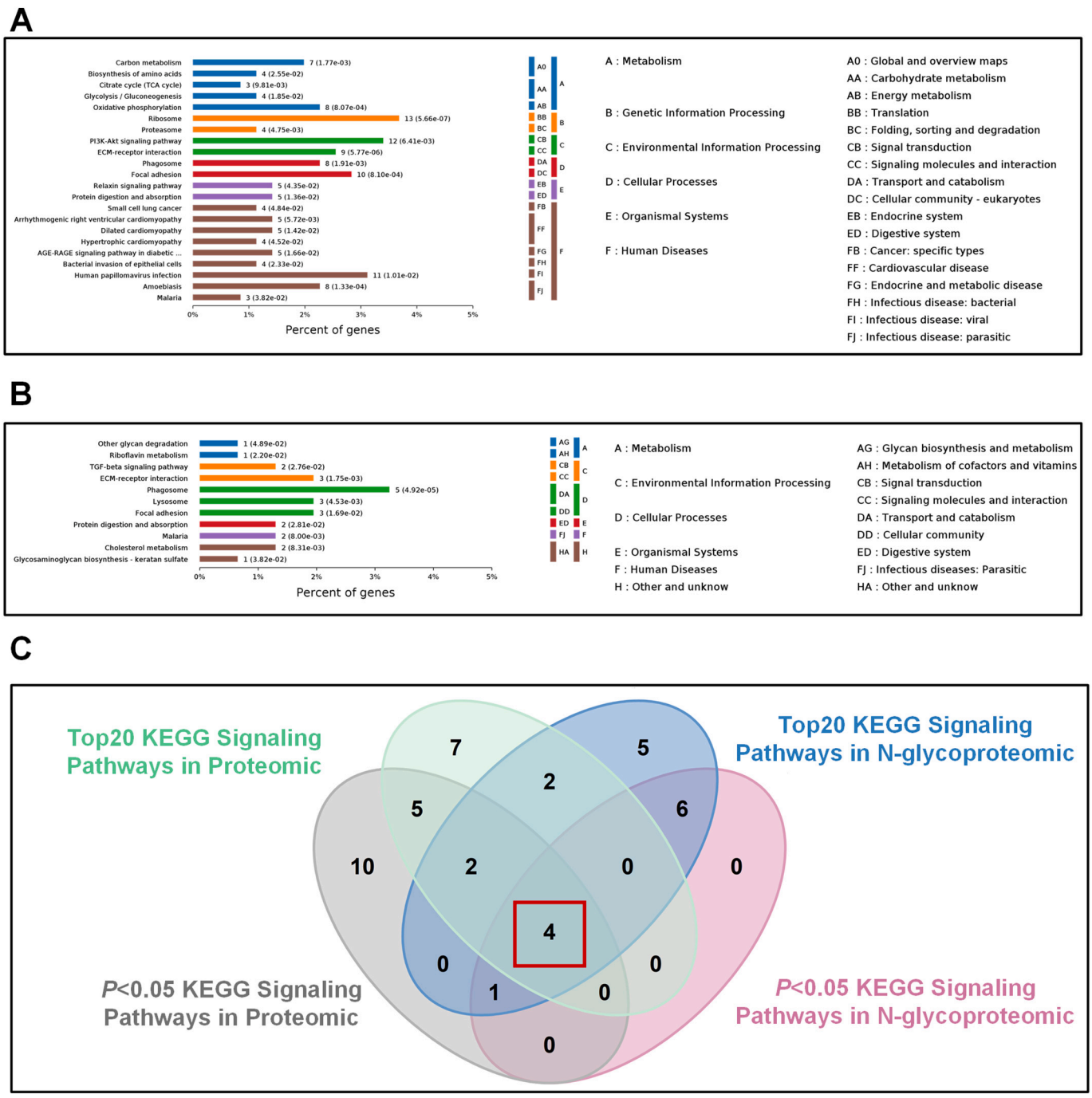



**Fig. 5.** KEGG pathways for differentially expressed proteins and N-glycoproteins in the cartilage of KBD and OA. (A) A list of proteins that were significantly associated with each category in proteomic KEGG pathway analysis. (B) A list of N-glycoproteins that were significantly associated with each category in N-glycoproteomic KEGG pathway analysis. (C) The number of top 20 KEGG signaling pathways and significant signaling pathways of proteomic and N-glycoproteomic results in KBD and OA patients.

fibroblast-like synoviocytes derived from OA patients [26].

The role of the N-glycosylated ITGB1 has been studied, for instance, for fibronectin-binding heterodimer it forms with α5 subunit [27]. In this heterodimer, the α5 subunit has 14 potential N-glycosylation sites and 12 in the β subunit [27]. N-glycosylation site numbers 3–5 of α5 subunit's β-propeller are essential for cell spreading and migration [28]. Three N-glycosylation sites in the β1 subunit, Asn-97, 212, and 269, are necessary for biological functions, such as cell spreading and dimer formation [27]. In the proteomic result of this study, ITGB1 was present in both KBD and OA subjects, but its expression level in KBD was lower than that in OA. In our study, two significant N-glycopeptides including four N-glycosylation sites were found: Asn-481, 520, 528, and 529. Asn-481 was located in peptide: CHEGN*GTFECGACR. Asn-520, 528, and 529 were located in peptides: KEN*SSEICSN*N*GECVCGQCVCR. The modification possibilities at the three sites of this peptide are 100% for Asn520, 5.8% for Asn528, and 94.2% for Asn529. These four N-glycosylated sites of ITGB1 were detected only in KBD. Asn481 and 520 are common N- glycosylation sites and have been recorded in UniProtKB. Therefore, it is speculated that the modification of these N-glycosylation sites may affect the generation of integrin dimers and play a role in the pathological process of KBD.

LRP1 is known as alpha-2-macroglobulin receptor (A2MR),

apolipoprotein E receptor (APOER), CD antigen (CD91). LRP1 endocytic receptor is involved in endocytosis and phagocytosis of apoptotic cells [29]. The activities of aggrecanase (the aggrecan proteoglycan degrading) and MMPs are highly significant in arthritis [30]. Normally, they are rapidly endocytosed and degraded by chondrocytes, but in OA the system is impaired by increased shedding of LRP1 [31]. This reduced LRP1–mediated endocytosis may underlie the increased ADAMTS-5 and MMP-13 activities in the disease [31]. In this study, LPR1 appeared to be present in equal contents in the proteomes of both KBD and OA, while the N-glycosylation site Asn-4125 was only found in KBD subjects. Therefore, it is proposed that this N-glycosylation in KBD patients may affect the protease level and activity through transmembrane transportation, and indicate the difference in pathogenesis between OA and KBD.

In proteomic studies, anoctamin (B4DZA5) was detected in both KBD and OA subjects. However, the N-glycosylation site of Asn-802 in anoctamin (B4DZA5) was found only in KBD. In UniProtKB, the status of anoctamin was unreviewed, while the similarity of the protein sequence between anoctamin and anoctamin-6 (ANO6) was 91.094%. ANO6 has been shown to participate in osteoclast formation in bone through phosphatidylserine exposure [32]. These studies suggest that ANO6 may be associated with bone development or pathology, which also is associated with KBD and OA. The specific N-glycosylation site Asn-802 in protein anoctamin may also indicate an association with the occurrence of KBD.

The protein D3DTX7 was down-regulated in the proteomics of KBD vs. OA but was found only in KBD subjects in N-glycoproteomic analysis. D3DTX7 corresponds to the gene COL1A1 and is currently unreviewed in UniProtKB. However, the amino acids of D3DTX7 are almost identical to the 579th amino acid sequence that was verified as COL1A1 protein P02452. The only difference is that the 1075th amino acid of P02452 is threonine (T), while the 496th amino acid of D3DTX7 is alanine (A). Asn-786 was found only in KBD in N-glycoproteomic analysis. N-glycosylation site Asn-786 of D3DTX7 corresponds to Asn-1365 of P02452. Asn-1365 happens to be the common N-glycosylation site of COL1A1. Previous studies have shown that this N-glycosylation site has been identified in breast cancer xenografts tissues [33]. N-glycosylation of the COL1A1 site also exists in human cerebrospinal fluid [34]. Therefore, we infer that the modification of COL1A1 by N-glycosylation may affect cartilage synthesis in KBD by influencing this process.

MXRA5 is related to matrix remodeling and cell-cell adhesion and belongs to a family of adhesion proteoglycan [35]. It was found as potential marker gene to define and influence cartilage/chondrocyte integrity and regeneration [36]. MXRA5 was found as significant N-glycoproteins (Asn-2007) in osteonecrosis of femoral head research when compared with the normal population [37]. In this study, we observed that the N-glycosylation site Asn-2500 was found only in KBD subjects. That N-glycosylation-modified MXRA5 maybe participate in the pathological mechanism of KBD.

DPP4, also called T-cell activation antigen (CD26), is a type II transmembrane glycoprotein with extracellular dipeptidyl peptidase proteolytic activity [38]. DPP4/CD26 plays a central role in protecting articular cartilage against invasion by synovial fibroblasts in rheumatoid arthritis [39]. It has been shown that the inhibition of DPP4 could ameliorate the progression of OA [40]. In this study, DPP4 was detected in both KBD and OA proteomic data, however only the Asn-520 site of KBD was modified in comparison to OA. This site is a common N-glycosylation site for DPP4 [41]. To our knowledge, there are no studies that have showen the relationship between the modification of the Asn529 site in KBD or bone and joint diseases till now. Therefore, the present study may clarify the different role of DPP4 in KBD.

CSPG4 plays a role in cell proliferation and migration that stimulates endothelial cell motility during microvascular morphogenesis. Although CSPG4 is not a major proteoglycan in cartilage, it has been noticed to be changed in arthritic diseases. Overexpression of lysyl oxidase like-2 amine oxidase increased mRNA levels of CSPG4 in OA chondrocytes [42]. Carbohydrate sulfotransferase 3 (CHST3) overexpression enhanced cartilage endplate-derived stem cells to repair intervertebral disc degeneration of rat nucleus pulposus via interaction with CSPG4 [43]. Proteomic analysis in the present study showed that the expression of CSPG4 in OA was higher than that in KBD. However, N-glycoproteomic analysis in the present study showed that only N-glycosylation sites Asn-2034 and 2040 were found in KBD subjects. UniProtKB showed that Asn-2034 and 2040 are common CSPG4 N-glycosylation modification sites. We did not find any published studies to explain the possible mechanism for the occurrence of these sites either. Therefore, our study may fill the gap in this field and provide help for the study of KBD occurrence.

CRLF1 is homologous with type I cytokine receptors [44]. It is reported that CRLF1 is expressed in osteoblasts and chondrocytes during endochondral ossification in mice, and may play a role in cartilage or bone formation [45]. In our study, CRLF1 was present in both KBD and OA proteomic analyses with no significant difference. However, in the N-glycoproteomic analysis, CRLF1 showed two significant N-glycosylation peptides, including four N-glycosylation sites. Asn-92, 102, and 104 occurred only in OA, Asn-168 appeared only in KBD. We speculated that different modification sites might influence the course of the disease.

GLG1, also called E-selectin ligand 1(ESL-1), is an important regulator of bone remodeling. Its modulation of TGF-β maturation is pivotal in the maintenance of a homeostatic bone microenvironment and for proper osteoblast–osteoclast coupling [46]. The present study showed that GLG1 was detected in both KBD and OA subjects by proteomic analysis. In N-glycoproteomic analysis, N-glycosylation site Asn-210 was found only in KBD subjects, while Asn-786 was found only in OA subjects. Our findings on the detection of GLG1 in both KBD and OA may provide a hint to understanding the N-glycosylation effect of these sites and suggest the different effects of N-glycosylation in the pathogenesis of OA and KBD.

In GO annotation for analysis, we found most of the CC and MF involved in both KBD and OA involve extracellular components. Among the CC, vesicle, membrane-bounded vesicle, and extracellular vesicle (EV) were noted in proteomics and glycosylated proteomics. Mesenchymal stem cells-derived EVs are capable of promoting tissue repair and regeneration in various diseases [47]. Their roles are particularly crucial when cartilage homeostasis is disrupted in OA with an imbalance in the catabolic and anticatabolic mediators. Human adipose-derived stem cells EVs are considered as a potential therapeutic approach in the treatment of OA. They not only promoted cell proliferation and migration but also maintained the chondrocyte matrix by increasing collagen II and decreasing MMPs [48]. EVs also can inhibit TNF-α-induced collagenase activity and promote cartilage regeneration in vitro [49]. The application of EVs in future KBD research may provide a new idea for the treatment and protection of KBD.

Common KEGG pathways include ECM-receptor interaction, focal adhesion, phagosome, protein digestion, and absorption. ITGB1 was involved in ECM-receptor interaction, focal adhesion, and phagosome. ITGB1 exists in the proteomics of KBD and OA, and the expression of ITGB1 in KBD is lower than that in OA. It is further proved that ITGB1 plays an important role in the research of KBD and OA.

Some limitations in the present study should be acknowledged. First, there is a difference in timing between proteomic and N-glycoproteomic analyses. The databases used in comparison analysis are Uniprot_human_156914_20170308.fasta and Uniprot_human_20190102_177661.fasta. Differences in databases may lead to the loss of a few proteins. Second, anoctamin (B4DZA5) protein has not been annotated in UniProtKB. We analyzed it only by looking for proteins that are highly similar to it. In future studies, the analyzed proteins should be annotated by UniProtKB. Third, the study lacked a validation part. This may affect the persuasiveness of results, but it also points out the direction for future research.

## 5. Conclusion

A total of 375 significantly different proteins and 278 N-glycosylation sites with 187 N-glycoproteins were identified in both KBD and OA. Proteins ITGB1, LRP1, AON6, COL1A1, MXRA5, DPP4, CSPG4 and their N-glycosylation sites were associated with the pathological process of KBD, yet CRLF1 and GLG1 were associated with both KBD and OA. GO annotation showed that EVs may play an important role in the occurrence of both KBD and OA. Key pathways in KBD and OA included ECM-receptor interaction, focal adhesion, phagosome, protein digestion, and absorption. This information provides substantial insight into the role of protein N-glycosylation in both KBD and OA, which also shows new ideas for future research on the pathogenesis of KBD and OA, as well as the other cartilage diseases.

## Ethics approval and consent to participate

All related analysis methods in the present study were approved by Ethics Committees at Xi'an Jiaotong University (No. 2017-196). All participants have signed informed consent for the study participation and publication of their clinical details and images before the study started.

## Consent for publication

Not applicable.

## Availability of data and materials

All data generated or analyzed in this study are included in this published article and its supplementary files.

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## CRediT authorship contribution statement

Yizhen Lyu: Conceptualization, Formal analysis, Investigation, Methodology, Writing - original draft. Huan Deng: Data curation, Formal analysis, Software, Roles/Writing - original draft. Chengjuan Qu: Conceptualization, Supervision, Writing - review & editing. Lichun Qiao: Formal analysis, Software, Writing - original draft. Xuan Liu: Conceptualization, Data curation, Writing - original draft. Xiang Xiao: Conceptualization, Writing - original draft. Jiaxin Liu: Formal analysis, Methodology. Ziwei Guo: Conceptualization, Investigation. Yan Zhao: Formal analysis, Software. Jing Han: Conceptualization, Funding acquisition, Project administration, Writing - review & editing. Mikko J. Lammi: Conceptualization, Investigation, Methodology, Writing - review & editing.

## Declaration of competing interest

All the authors declare no conflict of interest concerning the work. The authors claim that this manuscript is original and none of the material in the paper has been published or is under consideration for publication elsewhere.

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## Appendix A. Supplementary data

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